

NUNO MELO

Final-Year Software Engineering Student

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INTRODUCTION

- Final-year Software Engineering student at the University of Minho with hands-on experience in software development, systems, networking, databases, and collaborative academic projects. Through my studies, I have worked on projects involving network protocols, distributed systems, data processing, software architecture, and machine learning. I enjoy learning new technologies, solving technical problems, and turning ideas into practical solutions.

EDUCATION

Universidade do Minho

Bachelor in Software Engineering,

Braga, Portugal

Sep 2022 – Present

- **Courses:** Databases, Software Architecture, Computer Networks, Computer Communications, Information Systems Security
- **Expected graduation:** January 2027
- **Activities:** Semana da Engenharia Informática (SEI), Cesium Workshops

PROJECTS

Computer Communications – Network Protocols & Distributed Systems

Universidade do Minho

Nov 2025 – Jan 2026

- Designed and implemented application-layer communication protocols over UDP and TCP in a simulated satellite network using CORE.
- Implemented reliable UDP communication with acknowledgements, retransmissions, timeouts, and flow control.
- Developed a TCP-based telemetry service supporting concurrent clients and continuous monitoring data transmission.
- Tested protocol robustness under latency, packet loss, and packet duplication.

Data Processing & Query System

Universidade do Minho

Sep 2023 – Jan 2024

- Developed a modular C system for parsing, storing, and querying large datasets using hash tables and dynamic data structures.
- Implemented and tested batch and interactive query processing, including pagination and result management.
- Investigated performance bottlenecks and optimized data structures and query execution, reducing runtime to approximately 70s on large datasets.
- Conducted functional and performance testing across different hardware configurations and dataset sizes.

BitOK - Desktop Assembly

Universidade do Minho

Sep 2024 – Jan 2025

- Contributed to the planning and specification of an assembly-line management system through requirements analysis and validation.
- Participated in UML modeling and definition of the system's structural and behavioral architecture.
- Contributed to database design and interface prototyping before implementation.
- Collaborated with the team throughout development, contributing to problem-solving and system validation.

SKILLS, LANGUAGES

- **Languages:** Portuguese (Native speaker), English (Fluent)
- **Soft Skills:** Teamwork, Problem Solving, Communication, Critical Thinking
- **Programming Languages:** Python, C++, C, Java, C#
- **Tools & Platforms:** Git, Docker, Linux, GitHub, KNIME
- **Web & Software Development:** JavaScript, Vue.js, ASP.NET
- **Databases:** MySQL, SQL Server